

MARKET BASKET ANALYSIS REPORT

Operations Analytics-1: Market Basket Analysis Report

Algorithm: Apriori (Frequent Itemset Mining & Association Rules)

Objective: Identify statistically significant purchasing patterns to drive kitchen station synchronization and dynamic menu bundling.

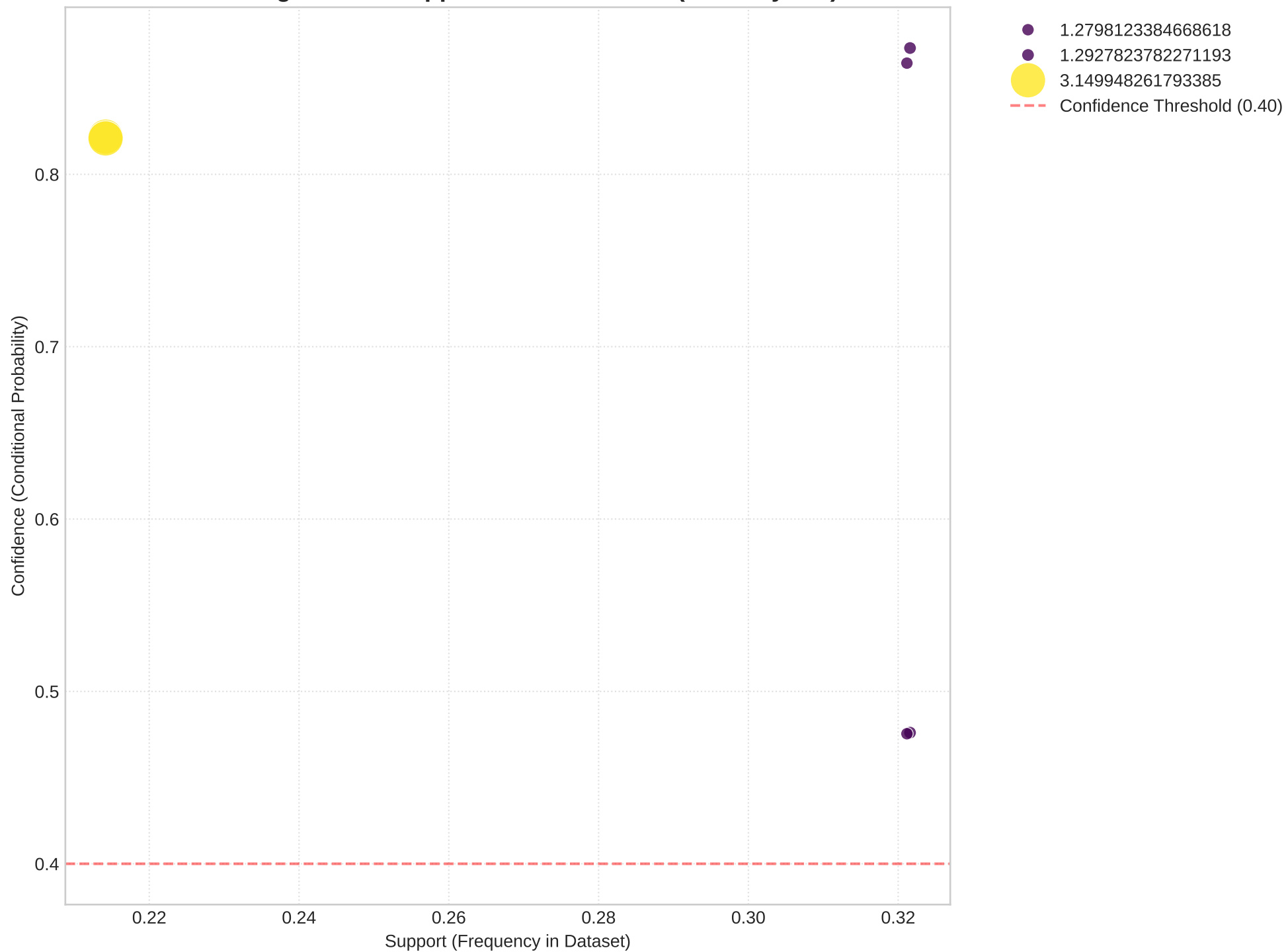
Top Frequent Itemsets (Support >= 5%)

support	itemsets
0.6755	Chicken Pepper Bbq Pizza (Medium)
0.3715	Virgin Mojito
0.3682	Cold Coffee
0.3216	Cold Coffee, Chicken Pepper Bbq Pizza (Medium)
0.3212	Virgin Mojito, Chicken Pepper Bbq Pizza (Medium)
0.2609	Bbq Fried Wings 6 Pcs.
0.2606	Chicken Lollipop 6pcs
0.2142	Chicken Lollipop 6pcs, Bbq Fried Wings 6 Pcs.
0.0658	Chilli Baby Corn (dry)

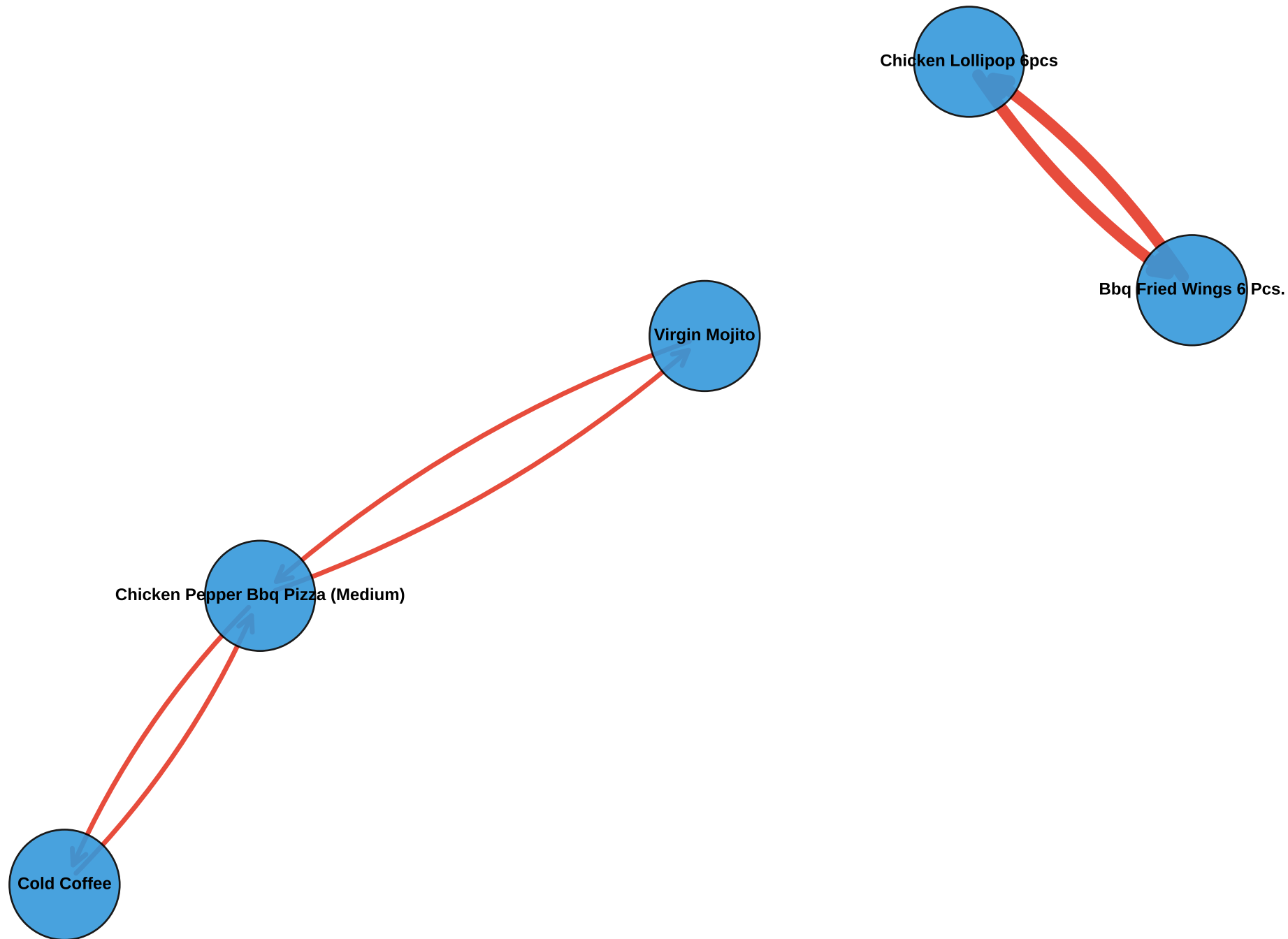
Top Strategic Association Rules

	Antecedent	Consequent	support	confidence	lift	leverage	conviction
	Chicken Lollipop 6p	Bbq Fried Wings 6 P	0.2142	0.8219	3.1499	0.1462	4.1492
	Bbq Fried Wings 6P	Chicken Lollipop 6p	0.2142	0.8208	3.1499	0.1462	4.1268
	Cold Coffee	Chicken Pepper Bbq Pizza	0.3216	0.8733	1.2928	0.0728	2.5607
Chicken Pepper Bbq Pizza		Cold Coffee	0.3216	0.4761	1.2928	0.0728	1.2058
	Virgin Mojito	Chicken Pepper Bbq Pizza	0.3212	0.8645	1.2798	0.0702	2.3951
Chicken Pepper Bbq Pizza		Virgin Mojito	0.3212	0.4755	1.2798	0.0702	1.1982

Rule Diagnostics: Support vs. Confidence (Sized by Lift)



Operations Network: Strategic Menu Associations



EXECUTIVE STRATEGY

Top Discovered Relationship:

[Chicken Lollipop 6pcs] ---> [Bbq Fried Wings 6 Pcs.]

Mathematical Proof:

- Support: 0.2142 (Occurs in 21.4% of all transactions)
- Confidence: 0.8219 (Given Antecedent, 82.2% chance of Consequent)
- Lift: 3.1499 (Customers are 3.15x more likely to buy this combination)

Strategic Implementation:

1. Kitchen Station Synchronization:

Because Lift > 1.2, the prep stations for these two items must be physically adjacent to minimize 'spaghetti flow' (wasted movement by chefs).

2. Dynamic Bundling (Commercial Operations):

Implement a POS-level prompt. Given the extreme conditional probability, bundling these items at a slight discount will drive guaranteed ticket volume without severely cannibalizing independent margins.